

Systems Architecture Document: Physics-Guided Thermal Infrared Spatial Super-Resolution Pipeline (200m to 100m)

1. Problem Statement

1.1 Context

Satellite-based Thermal Infrared (TIR) remote sensing is essential for monitoring global thermodynamic processes, including urban heat island (UHI) dynamics, evapotranspiration fluxes, agricultural water stress, and geothermal anomalies. However, due to the trade-offs in optical instrument design between a detector's signal-to-noise ratio (SNR) and spatial resolution, thermal sensors inherently possess lower spatial resolution than their Visible and Near-Infrared (VNIR) counterparts.

1.2 Specific Challenge

[cite_start]In the context of the Bharatiya Antriksh Hackathon (ISRO Problem Statement #10), we address the spatial constraints of Landsat 9's Thermal Infrared Sensor-2 (TIRS-2) Band 10 data[cite: 7]. [cite_start]While Band 10 is natively acquired at a 100 m spatial resolution, the standard Level-1 products distributed by space agencies are resampled onto a 30 m coordinate grid or scaled down to a 200 m baseline to match regional multi-spectral grids[cite: 9]. [cite_start]This resampling blurs sub-pixel thermal boundaries, limiting its utility for field-scale environmental modeling[cite: 93].

[cite_start]The primary objective is to develop an elegant, implementable, and mathematically rigorous machine learning pipeline that increases the spatial resolution of this degraded thermal imagery from 200 m to 100 m[cite: 90].

1.3 Core Objectives

- **Resolution Scaling:** Enhance the spatial resolution of TIRS-2 Band 10 data from a 200 m low-resolution (LR) baseline to an enhanced 100 m high-resolution (HR)

state ($2\times$ spatial upscaling factor)[cite: 7, 90].

- **Physical Consistency:** Ensure that the downscaling pipeline adheres to the law of conservation of energy, meaning the total radiant flux emitted within a 200 m parent pixel matches the integrated flux of its four constituent 100 m sub-pixels[cite: 95, 117].
 - **Artifact Suppression:** Suppress unphysical structural hallucinations, checkerboard artifacts, and edge blurs common in standard generative computer vision networks (e.g., standard SRGANs or unconstrained diffusion models)[cite: 94, 99].
 - **Boundary Sharpening:** Preserve the micro-thermal boundaries between adjacent land-cover classes (e.g., soil-water interfaces, urban footprint boundaries) by leveraging structural edge maps derived from synchronized VNIR surface emissivity signatures[cite: 25, 26, 42].
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2. Scientific Motivation

2.1 The Limits of Standard Interpolation

Traditional spatial upscaling methods, such as bilinear, bicubic, or Lanczos interpolation, rely on local pixel weight smoothing[cite: 93]. While computationally efficient, these mathematical operations function purely as low-pass filters. They smooth out sub-pixel transitions, alter local thermal gradients, and blur spatial features[cite: 93]. This structural blurring introduces uncertainty into downstream quantitative models, such as surface energy balance algorithms.

2.2 The Failures of Unconstrained Computer Vision Models

State-of-the-art super-resolution architectures from the computer vision domain (such as ESRGAN or conditional latent diffusion models) optimize for high perceptual quality and sharp edges by using adversarial or score-based losses[cite: 99, 100]. However, these networks operate without physical constraints[cite: 99]. When applied to scientific remote sensing datasets, they create realistic-looking but entirely hallucinated spatial thermal patterns[cite: 99]. These artifacts violate thermodynamic principles, distorting the true radiometric measurements collected by the spacecraft sensor[cite: 99].

2.3 The Hybrid Solution: Physics-Guided Super-Resolution

This pipeline bridges this gap by embedding remote sensing radiometry directly into a deep learning transformer architecture. Rather than feeding raw, uncalibrated

instrument digital counts directly to a neural network—which forces the weights to model sensor-specific electronic gains and atmospheric distortions [cite: 21]—we construct a dedicated **Radiometric Calibration and Feature Engineering Layer**. By transforming input data into physical invariants (such as Land Surface Temperature and Land Surface Emissivity) prior to upscaling, the model's parameters optimize for true surface thermodynamics rather than sensor-specific biases[cite: 23].

3. Physical Foundations & Sensor Physics Background

3.1 Thermal Infrared Radiometry

Every physical object with a thermodynamic temperature above absolute zero (0 K) continuously emits electromagnetic radiation. This behavior is governed by Planck's Law, which defines the spectral radiance spectral distribution of a perfect blackbody:

$$B_{\lambda}(\lambda, T) = \frac{2hc^2}{\lambda^5 \left(\exp \left(\frac{hc}{\lambda k_B T} \right) - 1 \right)}$$

Where:

- $h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$ (Planck's constant)
- $c = 2.99792458 \times 10^8 \text{ m/s}$ (Speed of light)
- $k_B = 1.380649 \times 10^{-23} \text{ J/K}$ (Boltzmann constant)
- λ = Electromagnetic wavelength (m)
- T = Absolute thermodynamic temperature (K)

Real-world surface features are not perfect blackbodies; they emit only a fraction of this theoretical maximum radiant energy[cite: 45]. This efficiency ratio is defined as the Land Surface Emissivity (ϵ), an intrinsic physical material property ranging from $0 \leq \epsilon < 1$ [cite: 18, 47]. The actual spectral radiance (L_{λ}) leaving a natural surface target is expressed as:

$$L_{\lambda} = \epsilon_{\lambda} B_{\lambda}(\lambda, T) + (1 - \epsilon_{\lambda}) L_{\text{atm}}^{\downarrow}$$

Where $L_{\text{atm}}^{\downarrow}$ represents the downward atmospheric thermal radiance reflecting off the terrain.

3.2 Radiometric Calibration Layer

The pipeline ingests raw Level-1 quantized 16-bit digital numbers (DN) from Landsat 9 TIRS-2 Band 10[cite: 7, 10]. We strip out detector-specific base gains by applying the

linear rescaling coefficients defined in the USGS Landsat 9 Data Users Handbook:

$$L_{\lambda} = M_L \cdot Q_{\text{cal}} + A_L$$

Where:

- Q_{cal} is the raw quantized input pixel tensor (DN) [cite: 8]
- M_L is the multiplicative band rescaling factor (`RADIANCE_MULT_BAND_10`) [cite: 31, 32]
- A_L is the additive band rescaling factor (`RADIANCE_ADD_BAND_10`) [cite: 32, 33]
- L_{λ} is the resulting Top-of-Atmosphere (TOA) spectral radiance in $\text{W} \cdot \text{m}^{-2} \cdot \text{sr}^{-1} \cdot \mu\text{m}^{-1}$ [cite: 33].

3.3 Temperature and Emissivity Extraction

By inverting Planck's Law under a blackbody assumption ($\epsilon = 1$), the pipeline converts TOA spectral radiance into an at-sensor Brightness Temperature (T_B) [cite: 35, 36]:

$$T_B = \frac{K_2}{\ln \left(\frac{K_1}{L_{\lambda}} + 1 \right)}$$

Where K_1 and K_2 are the standard pre-launch calibration constants for Landsat 9 Band 10 [cite: 38]:

- $K_1 = 774.89 \text{ Bowen} \cdot \text{m}^{-2} \cdot \text{sr}^{-1} \cdot \mu\text{m}^{-1}$ [cite: 39]
- $K_2 = 1321.08 \text{ K}$ [cite: 40]

Note: The structural matrix extraction formula is mathematically defined by the single inverse logarithmic denominator shown above [cite: 37].

Emissivity Estimation via Fractional Vegetation Cover (FVC)

To isolate true thermodynamic kinetic profiles from emissivity variations, we co-register complementary high-resolution visible red (ρ_{Red}) and near-infrared (ρ_{NIR}) channels [cite: 42]. First, the Normalized Difference Vegetation Index (NDVI) is computed [cite: 42]:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}}$$

The NDVI array is then mapped to the Fractional Vegetation Cover (FVC), which scales the pixel landscape composition between bare soil and closed green canopies [cite: 43]:

$$\text{FVC} = \left(\frac{\text{NDVI} - \text{NDVI}_{\text{soil}}}{\text{NDVI}_{\text{veg}} - \text{NDVI}_{\text{soil}}} \right)^2$$

Where $\text{NDV } I_{\text{soil}} = 0.05$ represents pure exposed soil, and $\text{NDV } I_{\text{veg}} = 0.70$ represents dense vegetation zones[cite: 44].

The Land Surface Emissivity (ϵ) map is extracted using the threshold-based vegetation allocation framework[cite: 46]:

$$\epsilon = \epsilon_{\text{veg}} \cdot \text{FVC} + \epsilon_{\text{soil}} \cdot (1 - \text{FVC}) + C_{\epsilon}$$

Where $\epsilon_{\text{veg}} = 0.99$, $\epsilon_{\text{soil}} = 0.97$, and C_{ϵ} is a cavity interaction factor modeling internal multi-reflections within rough canopy structural geometries[cite: 48]:

$$C_{\epsilon} = 0.005 \cdot \text{FVC} \cdot (1 - \text{FVC})$$

The localized true Land Surface Temperature (LST) is isolated via a single-channel radiative transfer inversion[cite: 50]:

$$\text{LST} = \frac{T_B}{1 + \left(\frac{\lambda \cdot T_B}{\rho} \right) \ln(\epsilon)}$$

Where $\lambda = 10.89 \mu\text{m}$ represents the effective center wavelength of TIRS-2 Band 10, and $\rho = \frac{h \cdot c}{\sigma_B} = 1.438 \times 10^{-2} \text{ m} \cdot \text{K}$ [cite: 52].

3.4 Physics-Guided Feature Engineering

Rather than passing a single thermal layer to our deep learning network, we construct a 3-channel physical feature block (Φ_{Feature})[cite: 11, 87]:

$$\Phi_{\text{Feature}} = [T_B \parallel \epsilon \parallel |\nabla T_B|] \in \mathbb{R}^{3 \times H_{\text{LR}} \times W_{\text{LR}}}$$

Where $|\nabla T_B|$ represents the spatial gradient magnitude of the brightness temperature field[cite: 87]:

$$|\nabla T_B| = \sqrt{\left(\frac{\partial T_B}{\partial x} \right)^2 + \left(\frac{\partial T_B}{\partial y} \right)^2}$$

Including Land Surface Emissivity (ϵ) provides the transformer backbone with explicit structural edge maps[cite: 25]. Because material composition boundaries (e.g., concrete infrastructure edges vs. water body shorelines) correlate with strict geometric features, the emissivity map acts as a spatial prior[cite: 25, 26]. This helps prevent edge drift or artificial smoothing during the super-resolution process[cite: 26].

4. Thermal Denoising Architecture

4.1 Justification for Hybrid Denoising

While satellite sensors encounter low-frequency stray-light ghosting anomalies, full multi-parameter iterative matrix inversions require out-of-field planetary look-angles and calibration parameters that are rarely available during a fast-paced hackathon pipeline execution[cite: 67, 68]. Conversely, completely unconstrained data-driven deep networks tend to misinterpret high-frequency stripe noise or sensor artifacts as true micro-thermal structures, magnifying these errors during upscaling[cite: 69].

This pipeline addresses this issue by integrating a fast, deterministic **Non-Local Means (NLM) / BM3D Hybrid Denoising Block** right before the super-resolution neural layers[cite: 70]. This approach clears out fixed-pattern detector noise while preserving sharp spatial thermal edges[cite: 70].

4.2 Mathematical Mechanics of the Denoising Layer

The denoising filter models the cleaned value of a target pixel x as a normalized weighted average of all pixels y within a defined search window Ω :

$$L_{\text{clean}}(x) = \frac{1}{Z(x)} \sum_{y \in \Omega} w(x, y) L_{\text{measured}}(y)$$

Where $Z(x)$ is the normalizing constant:

$$Z(x) = \sum_{y \in \Omega} w(x, y)$$

The similarity weights $w(x, y)$ depend on the Euclidean distance between the local intensity vectors of patches N_x and N_y centered at pixels x and y :

$$w(x, y) = \exp \left(- \frac{\|L_{\text{measured}}(N_x) - L_{\text{measured}}(N_y)\|_2^2}{h^2} \right)$$

Where h acts as a filtering parameter that controls the degree of noise smoothing. This step smooths out high-frequency striping artifacts across the microbolometer arrays while preserving valid thermal step transitions[cite: 83].

5. Super-Resolution Network Design: SwinIR

5.1 Architectural Selection Analysis

To choose the most reliable upscaling framework for scientific remote sensing deployment, we benchmarked four distinct neural network classes[cite: 97, 98]:

Model Architecture	Complexity	Structural Strengths	Vulnerabilities & Risks	Hackathon Implementation Viability
SRCNN	Low (~ 57K params) [cite: 99]	Low computational footprint; fast training and inference speeds[cite: 99].	Prone to edge-blurring; fails to capture complex structures[cite: 99].	Highly viable but provides poor structural results[cite: 99].
EDSR	Medium (~ 43M params)	Eliminates batch normalization layers to stabilize absolute intensity values.	High memory consumption; prone to localized spatial blurring.	Good alternative baseline.
ESRGAN	High (~ 16.7M params) [cite: 99]	Generates sharp high-frequency details using dense residual connections[cite: 99].	High risk of hallucinating unphysical thermal patterns[cite: 99].	Unstable; risky for scientific data validation[cite: 99].
Diffusion Models	Extreme (~ 850M params) [cite: 100]	Excellent detail synthesis and high realism[cite: 100].	Slow inference speeds; requires extensive compute resources[cite: 100].	Too slow for practical hackathon iteration cycles[cite: 100].
SwinIR (Selected)	High (~ 11.8M params) [cite: 99]	Captures long-range spatial dependencies using windowed self-attention maps [cite: 99, 100].	Requires careful multi-component loss tuning[cite: 99].	Optimal balance of structural clarity and physical accuracy [cite: 99].

5.2 SwinIR Detailed Architectural Pipeline

SwinIR processes input data through three main stages: shallow feature extraction, deep feature extraction via Residual Swin Transformer Blocks (RSTB), and a high-resolution image reconstruction module.

1. **Shallow Feature Extraction:** A 3×3 convolutional layer projects the input physical feature tensor $\Phi_{\text{Feature}} \in \mathbb{R}^{3 \times H_{\text{LR}} \times W_{\text{LR}}}$ into a stable intermediate latent embedding space $\mathbf{F}_0 \in \mathbb{R}^{C \times H_{\text{LR}} \times W_{\text{LR}}}$ [cite: 86, 87].
2. **Deep Feature Extraction (RSTB Modules):** The intermediate features pass through a sequence of six Residual Swin Transformer Blocks (RSTB). Each RSTB contains several Swin Transformer Layers (STL) that apply windowed Multi-Head Self-Attention (W-MSA) and Shifted Window Multi-Head Self-Attention (SW-MSA) [cite: 99, 100]. By computing attention within local, non-overlapping windows that shift between successive layers, the network models long-range spatial context across different land-cover boundaries while maintaining a manageable computational footprint [cite: 99, 100].
3. **High-Resolution Reconstruction:** After deep feature extraction, a final convolutional layer aggregates the structural embeddings, combines them with a shallow feature skip connection ($\mathbf{F}_0$), and passes the result to a **PixelShuffle sub-pixel convolution layer**. This upscaling module projects the feature maps into the final enhanced spatial matrix $\hat{\mathbf{T}}_{\text{HR}} \in \mathbb{R}^{1 \times H_{\text{HR}} \times W_{\text{HR}}}$ at a 100 m spatial resolution scale [cite: 89, 90].

6. Mathematical Formulations & Loss Function

Derivations

The SwinIR backbone optimization is governed by a multi-objective loss function that balances pixel-level accuracy with physical thermodynamic constraints [cite: 102, 103]:

$$\mathcal{L}_{\text{Total}} = \omega_1 \mathcal{L}_{\text{Charbonnier}} + \omega_2 \mathcal{L}_{\text{SSIM}} + \omega_3 \mathcal{L}_{\text{Huber-Edge}} + \omega_4 \mathcal{L}_{\text{FFT-Spectral}} + \omega_5 \mathcal{L}_{\text{Flux-Con}}$$

6.1 Pixel-Level Data Fidelity Loss (Charbonnier Loss)

Unlike standard Mean Squared Error ($\mathcal{L}_{\text{MSE}}$) losses, which over-penalize large outliers and produce blurred edges, we use the Charbonnier loss formulation to improve structural stability [cite: 105, 106]:

$$\mathcal{L}_{\text{Charbonnier}} = \frac{1}{H_{\text{HR}} W_{\text{HR}}} \sum_{x=1}^{H_{\text{HR}}} \sum_{y=1}^{W_{\text{HR}}} \sqrt{\left(\hat{\mathbf{T}}_{\text{HR}}(x, y) - \mathbf{T}_{\text{HR_GT}}(x, y) \right)^2 + \delta^2}$$

Where $\delta = 10^{-3}$ is a small stabilization constant [cite: 106].

6.2 Structural Similarity Index Loss

The structural loss enforces spatial layout consistency, ensuring that objects in the super-resolved thermal output remain structurally aligned with the ground truth[cite: 107]:

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{T}_{\text{HR}}, T_{\text{HR_GT}})$$

$$\text{SSIM}(A, B) = \frac{(2\mu_A\mu_B + c_1)(2\sigma_{AB} + c_2)}{(\mu_A^2 + \mu_B^2 + c_1)(\sigma_A^2 + \sigma_B^2 + c_2)}$$

6.3 Edge Preservation Theory (Huber-Edge Loss)

To prevent spatial blurring across different land-cover interfaces (e.g., land-water boundaries), we track horizontal and vertical gradient fields using a Huber loss constraint[cite: 109, 110]:

$$\mathcal{L}_{\text{Huber-Edge}} = \frac{1}{H_{\text{HR}} W_{\text{HR}}} \sum_{x,y} \left[\psi(\nabla_x \hat{T}_{\text{HR}} - \nabla_x T_{\text{HR_GT}}) + \psi(\nabla_y \hat{T}_{\text{HR}} - \nabla_y T_{\text{HR_GT}}) \right]$$

$$\text{Where } \psi(t) = \begin{cases} \frac{1}{2}t^2 & \text{if } |t| \leq \alpha \\ \alpha(|t| - \frac{1}{2}\alpha) & \text{otherwise} \end{cases} \quad (\alpha = 0.1)$$

6.4 Frequency Domain Consistency Theory (FFT-Spectral Loss)

To eliminate high-frequency checkerboard patterns and artifact blurring common in convolutional neural networks, the models are optimized in the spatial frequency domain using 2D Fast Fourier Transforms (FFT)[cite: 112, 113]:

$$\mathcal{L}_{\text{FFT-Spectral}} = \left\| \left| \mathcal{F}(\hat{T}_{\text{HR}}) \right| - \left| \mathcal{F}(T_{\text{HR_GT}}) \right| \right\|_1$$

Where $\mathcal{F}$ represents the 2D Fast Fourier Transform operator, outputting the absolute structural frequency amplitude spectrum map[cite: 115].

6.5 Flux Conservation Theory

To ensure strict thermodynamic accuracy, the super-resolved output must honor the conservation of total radiant energy[cite: 95, 116]. This means the integrated radiance across any four super-resolved 100 m sub-pixels must exactly match the measured radiance value of their original 200 m parent pixel[cite: 117]:

$$\mathcal{L}_{\text{Flux-Conservation}} = \frac{1}{H_{\text{LR}} W_{\text{LR}}} \sum_{i=1}^{H_{\text{LR}}} \sum_{j=1}^{W_{\text{LR}}} \left[\frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 \hat{L}_{\text{HR}}(2i+m, 2j+n) \right] - L_{\text{LR_Clean}}(i, j)$$

Where the downscaled physical radiance values are derived by passing the temperature predictions through the inverse Planck formulation[cite: 119]:

$$\hat{L}_{HR}(x, y) = \frac{K_1}{\exp\left(\frac{K_2}{\hat{T}_{HR}(x, y)}\right) - 1}$$

7. Training Strategy & Dataset Preparation

7.1 Dataset Generation Pipeline

Because native 100 m thermal data matched with perfect 200 m un-sampled baselines is limited, we generate our paired dataset using a downscaling down-sampling framework:

1. **High-Resolution Ground Truth Generation:** Collect native Landsat 9 TIRS-2 Band 10 scenes (100 m native pixel matrices), applying linear calibration parameters to yield our target T_{HR_GT} maps[cite: 7].
2. **Low-Resolution Input Degradation:** Construct paired input tensors (200 m) by downsampling the high-resolution images by a factor of 2 using an area-averaging pooling operation[cite: 90]. This simulates a true spatial footprint degradation while honoring flux conservation.
3. **Co-registered VNIR Integration:** Co-register Landsat 9 Operational Land Imager (OLI) Band 4 (Red) and Band 5 (NIR) scenes, downsampling them to match the 200 m input grid to extract our input Emissivity (ϵ) maps[cite: 42].

7.2 Data Augmentation

To prevent model overfitting and help the transformer learn orientation-invariant thermal boundaries, patches are augmented using a localized randomization loop:

- Random spatial rotations (90° , 180° , 270°).
- Horizontal and vertical axis reflections.
- Random cropping into overlapping 64×64 low-resolution input windows, mapping directly onto 128×128 high-resolution target coordinates.

7.3 Training Hyperparameters

- **Optimizer:** AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$).
- **Learning Rate Schedule:** Initial learning rate of 2×10^{-4} , decayed using a Cosine Annealing schedule down to a minimum threshold of 1×10^{-6} .
- **Batch Size:** 32 patches per training iteration step.

- **Hardware Profile:** Optimized for standard execution inside an active hackathon sandbox, utilizing an explicit single NVIDIA A100/RTX4090 GPU system container instance[cite: 99].
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8. Pipeline Verification Checkpoints & Evaluation Metrics

To verify pipeline performance at each stage, the system monitors four quantitative metrics across the generated output fields[cite: 149, 150]:

8.1 Peak Signal-to-Noise Ratio (PSNR)

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right)$$

Where $\text{MAX}_I = 340.0 \text{ K}$ represents the maximum expected operational range of terrestrial surface temperatures[cite: 55].

8.2 Structural Similarity Index Measure (SSIM)

Evaluates the reconstruction accuracy of spatial structural details, boundaries, and textures[cite: 153]:

$$\text{SSIM} = \frac{(2\mu_{\text{GT}}\mu_{\text{pred}} + c_1)(2\sigma_{\text{GT_pred}} + c_2)}{(\mu_{\text{GT}}^2 + \mu_{\text{pred}}^2 + c_1)(\sigma_{\text{GT}}^2 + \sigma_{\text{pred}}^2 + c_2)}$$

8.3 Root Mean Squared Error (RMSE) in Kelvin

Measures the absolute temperature deviation from the ground truth across the generated landscape[cite: 156]:

$$\text{RMSE}_{\text{Kelvin}} = \sqrt{\frac{1}{H_{\text{HR}} W_{\text{HR}}} \sum_{x=1}^{H_{\text{HR}}} \sum_{y=1}^{W_{\text{HR}}} \left(T_{\text{GT}}(x, y) - \hat{T}_{\text{HR}}(x, y) \right)^2}$$

8.4 Ablation Study Matrix Design

To evaluate the contribution of each module, the validation pipeline runs systematically through successive feature omissions[cite: 162]:

Configuration Case	Active Loss Functions	Emissivity Maps	Target PSNR (dB)	Target RMSE (K)
Baseline Upscale	Bicubic Interpolation Only [cite: 163]	Absent	31.24	2.41
Vanilla SwinIR [cite: 163]	$\mathcal{L}_{\text{Charbonnier}}$ Only	Absent	36.89	0.85
Edge Enhanced	$\mathcal{L}_{\text{Charbonnier}} + \mathcal{L}_{\text{Huber-Edge}}$	Included	38.42	0.62
Complete System	All Losses ($\mathcal{L}_{\text{Total}}$)	Included	41.15	0.34

9. Expected Results, Limitations, & Future Work

9.1 Expected Spatial Outputs

By utilizing a windowed attention backbone combined with an emissivity edge-preservation network, this architecture delivers crisp thermal boundaries across complex landscapes[cite: 26, 99]. Sharp boundaries like land-water interfaces or urban field footprints avoid spatial blurring or edge bleeding[cite: 25, 82]. The inclusion of a flux conservation loss ensures that the integrated sub-pixel temperature values match the absolute radiometry measured by the satellite[cite: 117].

9.2 Limitations

- **Cloud Mask Constraints:** The radiometric calibration layer requires clear-sky observations. If thin cirrus clouds or shadow boundaries cover a region, the input VNIR reflectance profiles skew, which corrupts the derived emissivity values and introduces artifacts into the downscaled temperature output.
- **Temporal Saturation:** This model processes static scenes independently. It does not account for temporal variations or diurnal heating cycles, which can cause minor performance degradation when analyzing rapidly changing thermal events like forest fire fronts.

9.3 Future Work

This pipeline provides a solid foundation for future development and deployment as a full production system:

- **Full PDE-Constrained PINN Integration:** Transitioning from empirical loss constraints to embedding the complete transient atmospheric thermal diffusion partial differential equations (IPDEs) directly into the network layers[cite: 95, 123].
- **Sensor-Specific Stray-Light Inversion Blocks:** Incorporating full planetary edge-annulus line integration arrays to dynamically remove optical ghosting artifacts using raw spacecraft engineering data[cite: 72, 74].
- **Physics-Guided Diffusion Models:** Developing conditional diffusion architectures that use flux conservation vectors as reverse-process guidance steps to generate sharp, physically accurate sub-pixel structures[cite: 100].
- **Multi-Temporal Super-Resolution:** Integrating recurrent long short-term memory (LSTM) networks or temporal cross-attention blocks to capture seasonal changes and surface moisture dynamics over time.
- **Thermal Foundation Models:** Pre-training large-scale transformer networks on global thermal datasets to build generalizable multi-sensor capabilities across Landsat, Sentinel-3, and upcoming high-resolution thermal tracking missions.